



The combined effects of ocean warming and ocean acidification on Pacific cod (*Gadus macrocephalus*) early life stages

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Received: 23 June 2023 / Accepted: 11 April 2024 / Published online: 28 April 2024

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Abstract

The eastern North Pacific is simultaneously experiencing ocean warming (OW) and ocean acidification (OA), which may negatively affect fish early life stages. Pacific cod (*Gadus macrocephalus*) is an economically and ecologically important species with demonstrated sensitivity to OW and OA, but their combined impacts are unknown. Through a ~9-week experiment, Pacific cod embryos and larvae were reared at one of six combinations of three temperatures (3, 6, 10 °C) and two CO₂ levels (ambient: ~360 µatm; high: ~1560 µatm) in a factorial design. Both embryonic and larval mortality were highest at the warmest temperature. Embryonic daily mortality rates were lower under elevated CO₂ and there was no effect of CO₂ level on larval daily mortality rates. Growth rates of young larvae (0 to 11 days post-hatch) were faster at warmer temperatures and at high CO₂ levels, but growth during the 11–28 days post-hatch interval increased by temperature alone. The condition of larvae decreased with age, but less markedly under high CO₂ levels. However, at 6 °C, fish incubated in ambient CO₂ remained in higher condition than fish in the high CO₂ treatment throughout the experiment. Overall, temperature had the greater influence on Pacific cod early life stages across each measurement endpoint, while CO₂ effects were more modest and inconsistent. Subtle developmental differences in larval Pacific cod could be magnified later in life and important in the context of recruitment. These results show the complexity of stage- and trait-specific responses to and value of investigating the combined effects of co-occurring climatic stressors.

Keywords Early life history · *Gadus macrocephalus* · Multistressor · Ocean acidification · Ocean warming · Pacific cod

Introduction

Anthropogenic climate change has already led to major changes in the physical and chemical properties of the ocean. Broadly, ocean warming (OW) is the result of increased atmospheric warming largely due to increases in atmospheric CO₂. The intensity of OW can vary with oceanographic dynamics such as those that can promote periods of abnormally warm ocean conditions (e.g., marine heatwaves).

Ocean acidification (OA) is directly caused by increased atmospheric CO₂, where a substantial fraction of atmospheric CO₂ dissolves into the surface ocean waters, raising oceanic CO₂ concentrations and lowering the pH through the dissociation of CO₂ in water (Caldeira and Wickett 2005). High-latitude ecosystems are more susceptible to OA because cold water has a higher gas solubility and ice melt introduces fresh water, which lowers the buffering capacity of the system (Fabry et al. 2009; Mathis et al. 2011). These ecosystems also support high primary productivity, which can alter OA dynamics within the system due to initial CO₂ uptake during phytoplankton blooms and subsequent release of CO₂ through respiration after the blooms (Fabry et al. 2009; Wang et al. 2021).

The eastern North Pacific Ocean, including the Gulf of Alaska and the Bering Sea, has experienced rising ocean temperatures (Hermann et al. 2016), which recently has been driven by a series of marine heatwaves and an intense El Niño event (Bond et al. 2015; Zador and Yasumiishi 2017). The North Pacific Ocean is expected to continue to warm

Responsible Editor: H.-O. Pörtner.

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(Scott et al. 2016), experience more frequent marine heatwaves (Cooley et al. 2022), and see further decreases in ocean pH (Pilcher et al. 2022). These trends are concerning because the Gulf of Alaska and Bering Sea support many important commercial fisheries that provide over half of the groundfish catch in the United States (Fissel et al. 2017). Investigating how these co-occurring environmental stressors may affect fishery species, particularly at sensitive early life stages, is a salient issue for understanding future recruitment in these fisheries (Shepherd and Cushing 1980; Houde 1987).

Because fish are ectotherms, OW can have profound effects on fish physiology, growth, and behavior (Clarke and Johnston 1999). An increase in body temperature associated with OW can elevate the baseline metabolic costs for organisms, limiting the available energy for other life sustaining processes such as swimming, feeding, and reproduction (Rijnsdorp et al. 2009). For adult fish, migration away from warmer regions is a common response (Pinsky et al. 2013), and it has been observed in Bering Sea fish assemblages after recent warming events (Stevenson and Lauth 2019). Fish early life stages (embryos, larvae) tend to be one of the most thermally sensitive ones in the life cycle (Dahlke et al. 2020), yet they are incapable of large-scale movements to avoid warming regions. Ocean warming can lead to a suite of effects in larval fish with many expected to have negative consequences including increased mortality (Koenker et al. 2018), smaller hatch size (Gobler et al. 2018; Villalobos et al. 2020), and higher frequency of malformations (Pimentel et al. 2014). Increased ocean temperature can also promote growth and quicker developmental rates (Laurel et al. 2016). While mortality may be lower at larger larval sizes via decreased size-dependent predation (Miller et al. 1988), faster growth requires higher food intake (Laurel et al. 2011) and increases the risk of starvation-induced mortality. Altogether, a majority of the responses to OW are negative which suggests that this stressor could reduce future larval recruitment into the adult population.

The effects of OA are more nuanced than the effects of OW. Adult and juvenile fish have advanced acid–base regulatory capacities (Melzner et al. 2009) and can increase bicarbonate in their bloodstream to buffer hypercapnia (Hamilton et al. 2019). These physiological processes increase fish resiliency to OA (Kunz et al. 2016), but negative effects from OA can still occur. At these later life stages, maintaining a disrupted acid–base balance may come at an energetic cost, potentially leading to negative downstream secondary effects (Heuer and Grosell 2014). Early life stages are less likely to have achieved ionoregulatory and buffering capabilities, which increases their sensitivity to OA (Melzner et al. 2009), yet some larval fish have demonstrated a capacity for acid–base regulation (Bignami et al. 2013;

Pimentel et al. 2014; Hamilton et al. 2019). The capability of larval fish to buffer elevated CO₂ is likely driven by differing species-specific physiology, leading to a suite of responses that include negative impacts on survival (Gobler et al. 2018) and hatching success (Baumann et al. 2022), positive influences on growth (Munday et al. 2009), and no apparent effects (Hurst et al. 2012; Perry et al. 2015). This range in species-specific responses to OA can also be driven by aspects of the experimental design (endpoints measured, life stage of focus, duration of exposure, etc.) and/or life-history traits such as adaption to specific environmental conditions (Baumann 2019).

Interactions among environmental stressors may occur under the combination of OW and OA (Baumann 2019). Broadly, combined OW and OA can decrease tolerance to either OW or OA as single stressors (Pörtner and Farrell 2008; Pörtner 2010). For example, embryonic survival in the Antarctic dragonfish (*Gymnodraco acuticeps*) was lower under OW conditions, and further decreased under combined OA and OW exposure (Flynn et al. 2015). Similarly, larval inland silverside (*Menidia beryllina*) were less resilient to OA near thermal limits (Gobler et al. 2018). These results suggest that interpreting the impacts of OW or OA alone may mask additional organismal responses, and studies focused on the interactions between the co-occurring stressors can improve the predictive capacity from experimental studies that broadly represent future ocean conditions.

Throughout the Gulf of Alaska and Bering Sea, Pacific cod (*Gadus macrocephalus*) support the second largest groundfish fishery in Alaska with average landings valued in excess of \$400 M (wholesale value 2009–2019; Fissel et al. 2017). The life cycle of Pacific cod is complex, such that life stages are spatially separated leading to different thermal requirements based on age. The eggs are semi-demersal (Mecklenburg et al. 2002), and after hatching, larvae are pelagic inhabiting surface waters (Hurst et al. 2009). Pacific cod larvae hatch with small yolk reserves and begin feeding soon after hatch (Laurel et al. 2008, 2011). The effects of OW on Pacific cod are better understood than the effects of OA. The early life stages are sensitive to OW in terms of hatch success (Laurel and Rogers 2020), development and size-at-age (Laurel et al. 2008). Indeed, the recent heatwaves (2014–2016; 2019) led to a dramatic decrease in the Gulf of Alaska population of Pacific cod (Barbeaux et al. 2018, 2020), potentially driven by the loss of suitable spawning habitat (Laurel and Rogers 2020). In the only OA study on Pacific cod to date, Hurst et al. (2019) found that OA negatively impacted Pacific cod at an earlier life stage (2 weeks post-hatch), but as fish aged (5 weeks post-hatch), the OA response had reversed and become positive. There have yet to be interactive studies focused on the combined effects of OW and OA on Pacific cod.

This study investigated the interactive effects of OW and OA on Pacific cod embryos and larvae by rearing fish in a

factorial experimental design at three temperatures crossed with two CO₂ levels. The range of temperatures included cold (3 °C), mid (6 °C) and warm (10 °C) treatments, while an ambient and high (~1500 µatm) treatment were used for the CO₂ levels. Collectively, these treatments spanned current and future ocean conditions within the Gulf of Alaska and Bering Sea regions of the eastern North Pacific Ocean (Mathis et al. 2015; Hermann et al. 2019; Pilcher et al. 2022). Whole-animal traits, including mortality, growth, and condition, were measured throughout both the embryo and larval life stages. This design allowed evaluation of two interrelated goals: 1) provide additional insights on the responses of Pacific cod to OA and 2) assess how the thermal responses of Pacific cod may be altered by concurrent OA conditions.

Materials and methods

Animal collection and husbandry

Adult Pacific cod were collected aboard commercial fishing vessels using baited fishing pots northeast of Kodiak Island, AK, USA, in April 2022 during the spawning season. Once aboard, fish sex and spawning status were determined through the expression of either eggs or milt, and ripe fish were strip spawned to collect gametes. In total, eggs from two females were fertilized with a mixture of milt from three different males and the maternal half-siblings were held separately. Fertilized embryos from each egg batch were distributed across a series of 4 to 6 insulated 2-L bottles chilled to ~4 °C, and shipped to the National Marine Fisheries Service laboratory in Newport, OR where they arrived 24 h after fertilization. Upon arrival, the quality of the embryos from each batch was visually inspected with a dissecting microscope. During this initial inspection, the embryos from one of the two females was deemed to not be viable, likely due to the eggs not being fully hydrated prior to fertilization, and were not used in the study.

Pacific cod embryos and larvae were reared in cylindrical 50-L tanks (experimental tanks) with black walls and a light grey conical bottom. Water was provided through a flow-through upwelling system to a volume of 40 L per tank and a flow rate of 0.5 L min⁻¹. Pacific cod embryos are demersal and negatively buoyant. To provide an enclosed platform for the embryos to settle throughout their development, incubation baskets (5 L; 30 cm × 22 cm × 8 cm, L × D × H) were placed inside each experimental tank and floated at the surface (embryos were fully submerged). Embryos were patchily distributed in a single layer covering ~75% of the bottom of the basket (density in basket: ~2,100 embryos L⁻¹), which received direct inflow of treatment water at 100 mL min⁻¹. Embryos were then slowly acclimated

to experimental conditions over a period of 3 d. When hatching was complete, yolk-sac larvae were counted and then transferred from the incubation baskets to the same experimental tank, at which point food was immediately introduced. Larvae were provided live rotifers (*Brachionus plicatilis*) twice daily at a density of 5 rotifers mL⁻¹ with green water (RotiGreen Nanno; Instant Algae). Throughout the duration of the experiment, lights were kept at a 12:12 h schedule. Salinity, dissolved oxygen, and water quality were monitored throughout the experiment while temperature and pH were regulated for each treatment condition (see below).

Experimental set-up

Pacific cod embryos and larvae were reared at three experimental temperatures (3, 6, and 10 °C) and two CO₂ conditions (ambient and high [~1560 µatm]) in a full factorial design resulting in six treatment combinations and replicated four times (24 experimental tanks total). The temperature treatments were achieved by manually mixing ambient and chilled water. Ambient water was from Yaquina Bay, OR (average temperature ~11 °C) and the chilled water was from the same ambient water that had been cooled using a glycol cooling system (average temperature ~2 °C). The high CO₂ treatments were achieved through controlled bubbling of CO₂ into three of the header tanks (one per temperature treatment). Each high CO₂ header tank was continuously monitored with a Durafet III pH probe (Honeywell) connected to a dual input analytical analyzer (Honeywell) that maintained a predetermined pH target (target pH ~7.3) through controlled injection of CO₂ gas. Each header tank was aerated to maintain mixing and dissolved oxygen levels, and supplied water to four replicate experimental rearing tanks. Throughout the experiment, temperature and pH were monitored in the experimental tanks daily. Once per week, water samples were taken from experimental tanks, fixed with mercuric chloride (HgCl₂), and sent to the Ocean Acidification Research Center (University of Alaska at Fairbanks). Here, components of the carbonate system, total alkalinity (TA) and dissolved inorganic carbon (DIC), were measured using an AIRICA (Automated InfraRed Inorganic Carbon Analyzer) and VINDTA 3C (Versatile Instrument for the Determination of Total dissolved inorganic carbon and Alkalinity). These instruments were calibrated using Certified Reference Materials (CRMs) from the Dickenson Laboratory at the Scripps Institute of Oceanography (Batch 194), with a mean deviation from CRM values at ±1.40 µmol kg⁻¹ for DIC and ±2.00 µmol kg⁻¹ for TA. This information, along with the measured temperature and salinity, was used to calculate the corresponding pH (seawater scale) and CO₂ of the water using the package SeaCarb in R (Gattuso et al. 2021) using

the dissociation constants of Lueker et al. (2002). The water chemistry of the system throughout the experiment is reported in Table 1.

Embryonic and larval measurements

Pacific cod embryo hatching occurred across several days, the start and duration of which are temperature dependent (duration ~6–12 d from 2–8 °C; Laurel et al. 2008). To standardize, the end of hatching was designated as 0 days post-hatch (DPH) and determined through visual inspection of the number of remaining viable embryos in the baskets. At this time point, the number of hatched larvae were counted and the embryonic instantaneous daily mortality rate was calculated as the difference between the natural log of the number of hatched fish and natural log of the estimated initial embryo number divided by the time to hatch. After counting, a subset of 15 fish per tank were removed for measurement to provide length-, weight-at-hatch and yolk area. Individual larva were anesthetized with MS-222 and photographed with a dissecting microscope (3.2× magnification) to obtain measurements of standard length (SL; mm), myotome height at the anus (MH; mm), and yolk area (mm²) with image processing software (Image J). Dry weights (DW; mg) were obtained by pooling 5 larvae, rinsing with ammonium formate, and drying them on pre-weighed foils for 24 h at 60 °C. These traits (except yolk area) were repeatedly measured throughout the rest of the experiment on subsamples of ~15 fish per tank at 11, 21, 28, and 36 DPH. Due to varying mortality rates across temperature treatments, experiments were ended at 11, 36, and 28 DPH for 10, 6, and 3 °C treatments, respectively. The remaining larvae were counted to calculate a larval instantaneous daily mortality rate as described above for embryos.

The SL, MH, and DW data were used to calculate length- and mass-based growth rates as well as two condition factors. The length-based growth (G_L ; mm day⁻¹) was calculated as the change in SL over a specific time period while

the mass specific growth rate (G_M ; % day⁻¹) was calculated as the difference in the natural log of DW across a specific time period and multiplied by 100. Growth rate was calculated across two age intervals: period 1 (0–11 DPH) for all three temperature treatments, and period 2 (11–28 DPH) for 3 and 6 °C treatments. The morphometric-based condition factor, K_{MH} , was calculated as the deviation from measured to expected MH based on a quadratic fit between MH and SL for all fish (SI Fig. 1). The weight-based condition factor, K_{DW} , was defined as the measured DW divided by the predicted DW based on a broken stick regression of log₁₀(DW) and log₁₀(SL) (SI Fig. 2).

Data analysis

Data were analyzed using either Generalized Linear Models (GLMs) or Generalized Linear Mixed Effects Models (GLMMs) to assess the effect of temperature (3-level factor), CO₂ (2-level factor), and their interaction on embryonic and larval mortality rates, length-, weight, and yolk area-at-hatch, and growth rates (G_L , G_M). Condition was analyzed with age (DPH) as an additional continuous predictor variable, with the interactions of temperature and CO₂ level. GLMs were used for response variables measured at the tank level (e.g., mortality rates, growth rates). GLMMs were used for response variables where individual fish measurements were available, and included tank as a random effect. Random effects were included in each GLMM regardless of the amount of variation explained by the random effect to maintain statistical independence (Barr et al. 2013). Appropriate models were identified through comparisons of conditional R²'s, AIC scores, and visually by assessing the residuals. For K_{DW} , because larval fish were pooled to obtain dry weights, the number of observations within a tank were too few to fit data with a mixed-effects structure. Therefore, a GLM was used with tank mean as the level of observation. Weight-at-hatch, larval mortality rate, G_L , G_M , K_{MH} and K_{DW} were fit with a Gaussian distribution family. Embryonic mortality rate, length-at-hatch, and yolk area were fit with a Gamma

Table 1 Seawater carbonate chemistry during Pacific cod experimental rearing

CO ₂ Level	Temp Level	pH _{SWS}	pCO ₂	°C	Salinity	DIC (mol/kg)	TA (mol/kg)
Amb	3	8.12 ± 0.07	297.20 ± 65.51	3.03 ± 0.39	30.21 ± 1.11	1936.76 ± 76.24	2074.63 ± 65.73
Amb	6	8.03 ± 0.11	394.28 ± 132.84	6.14 ± 0.29	29.84 ± 1.48	1943.38 ± 105.62	2063.08 ± 84.73
Amb	10	8.00 ± 0.03	409.60 ± 36.75	10.22 ± 0.24	29.63 ± 0.67	1911.35 ± 22.43	2040.80 ± 21.72
High	3	7.48 ± 0.09	1434.95 ± 304.70	3.10 ± 0.18	29.77 ± 1.30	2093.99 ± 76.91	2050.55 ± 68.72
High	6	7.47 ± 0.03	1480.80 ± 126.37	6.04 ± 0.51	29.36 ± 1.31	2078.14 ± 81.47	2042.54 ± 84.58
High	10	7.40 ± 0.06	1780.01 ± 245.48	10.20 ± 0.37	29.63 ± 0.67	2076.27 ± 24.75	2038.04 ± 22.15

Values are presented as mean ± SD. Temperature and salinity were measured daily, and dissolved inorganic carbon (DIC) and total alkalinity (TA) were measured from preserved bottle samples that were collected weekly. pH_{SWS} and pCO₂ were calculated from temperature, salinity, DIC and TA

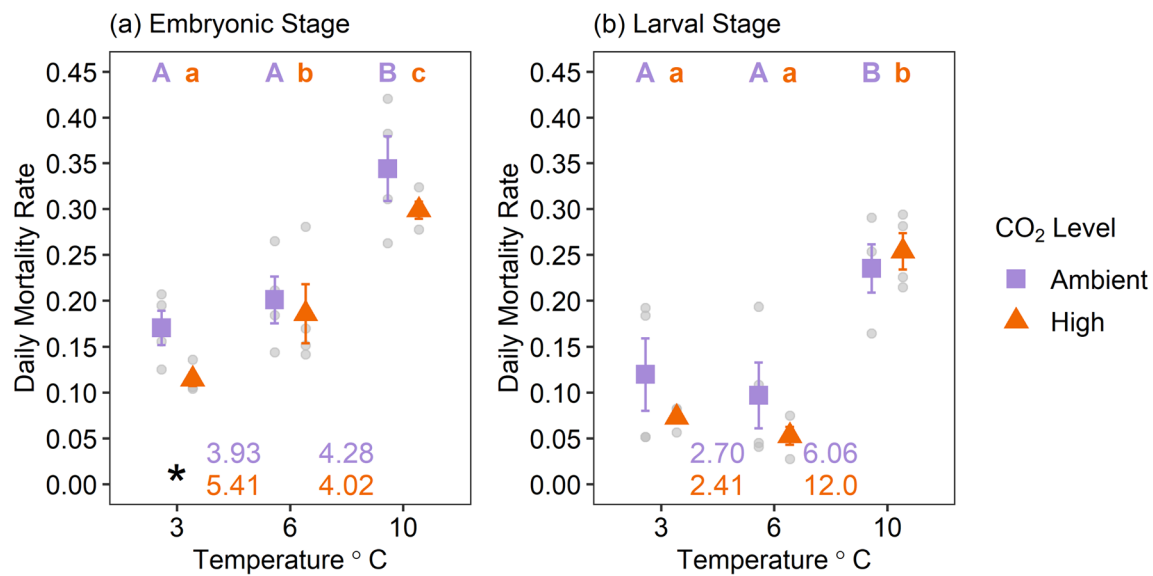


Fig. 1 Embryonic (a) and larval (b) daily mortality rates across temperature treatment with CO₂ level designated as ambient (purple square) and high (orange triangle). Values are presented as mean $\pm$ S.E. and overlain atop the tank-level data (grey points; $n=4$ per treatment). Significant differences between temperature treatments within a specific CO₂ treatment are depicted by different let-

ters at the top (upper case: ambient CO₂; lower case: high CO₂). The * indicates a significant difference between CO₂ level within a temperature treatment. The numbers in between temperature treatments indicate the Q₁₀ value between each temperature interval (3–6; 6–10) shown for ambient and high CO₂ as the top value (purple) and bottom value (orange), respectively

distribution with an inverse, log and identity link, respectively. To improve model fit, data were reflected and then log₁₀-transformed for length-at-hatch, and square root transformed for yolk area. After individual GLM or GLMMs were developed, the significance of overall treatment effects were analyzed using Wald Chi-square tests (Type III sums of squares ANOVA) and, if treatment effects were observed, pair-wise differences were assessed through a Tukey's post-hoc test. Where appropriate, data were also interpreted through Q₁₀ values (rate change) to express the temperature dependence of a process within a CO₂ treatment across the experimental temperature range. Significance level was set at an alpha of 0.05, and all analyses were performed in R (Version 4.1.2; R Core Team 2021).

Results

Mortality rates

Daily mortality rates of embryos were affected by temperature ($X_{24}=22.908$, $p<0.001$; Table 2), and post-hoc comparisons indicated that daily mortality rates were highest at 10 °C ($p<0.01$). While the duration to hatch was shortest for 10 °C (6 d) compared to at 6 °C (8 d) and 3 °C (12 d), the higher daily mortality rate of Pacific cod embryos at 10 °C resulted in a lower hatch rate than observed at 3 and 6 °C. The effect of CO₂ on embryonic daily mortality rates led to lower daily

mortality rates under high CO₂ conditions ($X_{24}=6.474$, $p<0.05$; Table 2). Post-hoc comparisons indicated this effect was significant in the 3 °C group. While there was no significant interaction between temperature and CO₂ in the full model ($X_{24}=4.070$, $p>0.05$; Table 2), the increase in daily mortality rates from 3 to 6 °C in the high CO₂ treatment was larger than seen in the ambient CO₂ treatment, and apparent in the comparison of Q₁₀ values between 3 and 6 °C within CO₂ treatments (ambient=3.93, high=5.41; Fig. 1a).

Temperature also had an independent effect on larval mortality rate ($X_{24}=16.234$, $p<0.001$; Table 2) with higher daily mortality rates observed at 10 °C ($p<0.01$; Fig. 1b) than at 3 and 6 °C. The difference in Q₁₀ values between the 3 to 6 °C interval (Q₁₀: 2.41–2.70) and 6 to 10 °C interval (Q₁₀: 6.06–12.0) also reflected the much higher mortality rates at 10 °C. There was no effect of CO₂ level ($X_{24}=1.625$, $p>0.05$; Table 2) or the interaction between temperature and CO₂ on larval mortality rate ($X_{24}=2.037$, $p>0.05$; Table 2). Although larval mortality rates were standardized based on the day the experiment ended for each temperature treatment, survival to a back-calculated common degree day (60 degree days) also reflected the higher survival seen at 6 °C (52% of fish) than at 3 (21% of fish) and 10 °C (24% of fish).

Size characteristics of larvae at hatching

Time to hatch differed across the temperature treatments where first hatchers were seen 19–20, 13–14, and 10–11 days

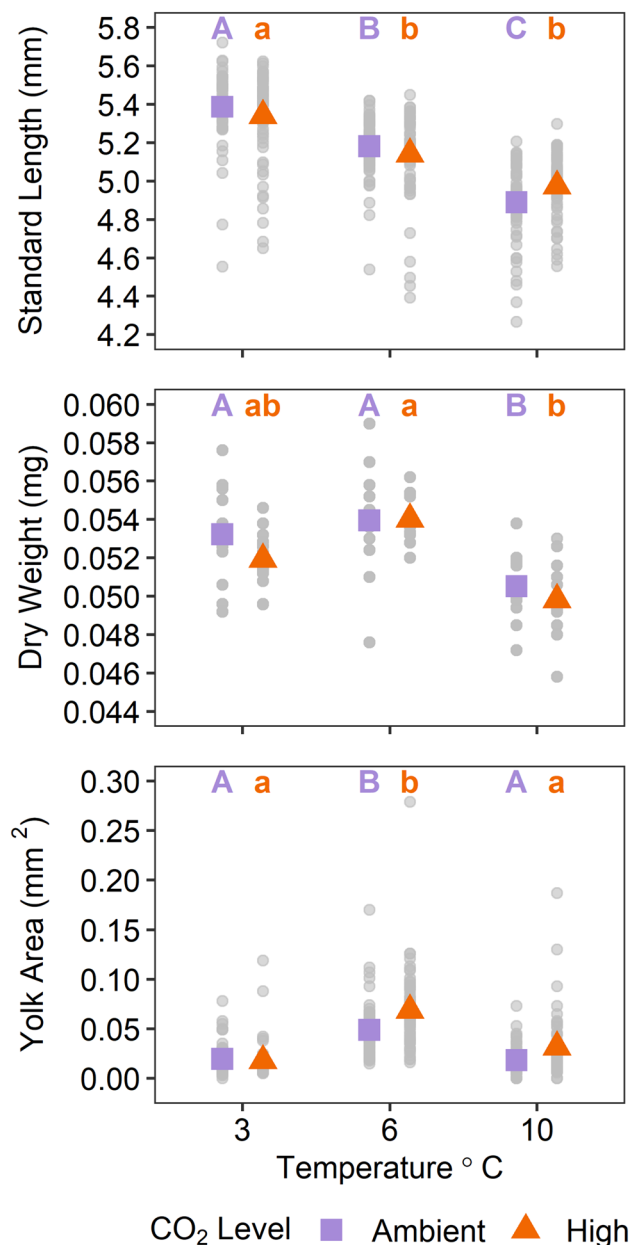


Fig. 2 Length-, weight-, and yolk area-at-hatch across temperature treatment with CO₂ level designated as ambient (purple square) and high (orange triangle). Values are presented as mean $\pm$ S.E. and overlain atop individual fish data (grey points; length: $n=52-67$, weight: $n=10-13$, yolk area= $52-67$ per treatment). Significant differences between temperature treatments within a specific CO₂ treatment are depicted by different letters at the top (upper case: ambient CO₂; lower case: high CO₂)

post fertilization in 3, 6, and 10°, respectively. There was no difference in time to hatch among CO₂ treatments. Length-, weight-, and yolk area-at-hatch were affected by temperature (length: $X_{343}=51.219$; weight: $X_{69}=13.504$; yolk area: $X_{330}=23.812$; all $p<0.001$; Table 3). Length-at-hatch increased with decreasing temperature (Fig. 2a), with Pacific

Table 2 Analysis of variance (Wald-Chi square tests) of embryonic and larval daily mortality rates

	Df	Embryonic		Larval	
		X ²	P-value	X ²	P-value
Temperature	2	22.908	<0.001	16.234	<0.001
CO ₂	1	6.474	0.011	1.625	0.202
Temperature $\times$ CO ₂	2	4.070	0.131	2.037	0.361

Values in **bold** indicate a significant difference (p -value <0.05). Full GLM coefficients are in SI Table 1

cod length-at-hatch 5% and 9% shorter at 6 and 10 °C, respectively, than at 3 °C. The weight-at-hatch was ~7% lighter in fish incubated at 10 °C than fish incubated at 3 and 6 °C (Fig. 2b). While fish at 6 °C hatched slightly heavier than at 3 °C, there was no significant difference between these two groups (Fig. 2b). Yolk area was largest at 6 °C compared to at 3 and 10 °C (Fig. 2c), which may have contributed to the increased weight-at-hatch. For length-, weight-, and yolk area-at-hatch, there was no effect of CO₂ level (length: $X_{343}=1.390$; weight: $X_{69}=1.806$; yolk area: $X_{330}=0.056$; all $p>0.05$; Table 3) or the interaction between temperature and CO₂ level (length: $X_{343}=1.834$; weight: $X_{69}=1.004$; yolk area: $X_{330}=3.048$; all $p>0.05$; Table 3). While there was no effect of CO₂ on yolk area, the post-hoc test indicated near significance between CO₂ treatments at 6 and 10 °C (6 °C: $p=0.0635$; 10 °C: $p=0.0734$), trending towards larger yolk area in the elevated CO₂ treatment. The random effect of tank had minimal effect on the results (SI Table 1).

Growth rates

Both G_L and G_M responded to temperature and CO₂ in a similar manner. Temperature had a direct effect on G_L and G_M for age intervals period 1 (0–11 DPH; G_L : $X_{24}=19.868$; G_M : $X_{24}=15.738$; all $p<0.001$; Table 4) and period 2 (11–28 DPH; G_L : $X_{16}=35.706$; G_M : $X_{16}=18.539$; all $p<0.001$; Table 4) with higher temperatures leading to faster growth rates (Fig. 3). However, the temperature effect on growth was more apparent at the second age interval (~2 $\times$ higher between 3 and 6 °C; Fig. 3a & c). There was an effect of high CO₂ during period 1 for both G_L ($X_{24}=5.681$, $p<0.05$; Table 4) and G_M ($X_{24}=4.493$, $p<0.05$; Table 4) where fish at elevated CO₂ grew faster than fish at ambient CO₂ (Fig. 3a & b). This trend was strongest at 3 °C, where Pacific cod incubated in the high CO₂ treatment grew twice as fast as those incubated in the ambient CO₂ treatment (Fig. 3a & b). A greater influence of elevated CO₂ on growth at 3 °C compared to at 6 and 10 °C was also reflected in the difference in Q_{10} values within CO₂ treatment between 3 and 6 °C for G_L (ambient = 6.49; high = 4.32) and G_M

Table 3 Analysis of variance (Wald-Chi square tests) of measurements at hatch

	Df	Length		Weight		Yolk Area	
		X ²	P-value	X ²	P-value	X ²	P-Value
Temperature	2	51.219	< 0.001	13.504	< 0.001	23.812	< 0.001
CO ₂	1	1.390	0.238	1.806	0.179	0.056	0.814
Temperature × CO ₂	2	1.834	0.399	1.004	0.605	3.048	0.218

Values in **bold** indicate a significant difference (p-value < 0.05). Full GLMM coefficients are in SI Table 1

Table 4 Analysis of variance (Wald-Chi square tests) of larval growth rates

	Df	G _L				G _M			
		P1		P2		P1		P2	
		X ²	P-value	X ²	P-value	X ²	P-value	X ²	P-value
Temperature	2	19.868	< 0.001	35.706	< 0.001	15.738	< 0.001	18.539	< 0.001
CO ₂	1	5.681	0.017	0.563	0.453	4.493	0.034	0.492	0.483
Temperature × CO ₂	2	1.016	0.602	0.101	0.751	0.477	0.788	0.582	0.446

Values in **bold** indicate a significant difference (p-value < 0.05). Full GLM coefficients are in SI Table 1. G_L length-based growth. G_M mass-based growth. P1 period 1 (0–11 DPH). P2 period 2 (11–28 DPH)

(ambient = 25.0; high = 5.51), which contrasted to similar Q₁₀ values between 6 and 10 °C in both CO₂ treatments (3.18–3.90). By period 2, there was no effect of CO₂ (G_L: X16 = 0.563; G_M: X16 = 0.492; all p > 0.05; Table 4) or the interaction between temperature and CO₂ (G_L: X16 = 0.101; G_M: X16 = 0.582; all p > 0.05; Table 4).

Condition

K_{MH} was significantly affected by temperature (X1181 = 11.958, p < 0.01; Table 5) and DPH (X1181 = 15.135, p < 0.001; Table 5), and the interactions between temperature and DPH (X1181 = 15.620, p < 0.001; Table 5) and temperature, CO₂ and DPH (X1181 = 9.050, p < 0.05; Table 5). At 0 DPH, fish incubated at 3 °C had significantly lower K_{MH} than at 6 and 10 °C (p < 0.05; SI Table 1). Generally, as fish aged, K_{MH} decreased, but the rate of decline varied among temperature and CO₂ treatments (Fig. 4). At 3 °C, there was a faster decrease in K_{MH} in the ambient compared to the high CO₂ treatment (slope = ambient: -0.0006 vs high: -0.0003; SI Table 1). At 6 °C, K_{MH} remained constant in the ambient CO₂ treatment but decreased in the high CO₂ treatment (slope = ambient: 0 vs high: -0.0005; SI Table 1). Finally, at 10 °C, K_{MH} decreased similarly in both the ambient and high CO₂ treatments (slope = ambient: -0.0011 vs high: -0.0013; SI Table 1). Among these relationships, the change in K_{MH} with age at 6 °C in ambient CO₂ was significantly different than that at 3 °C in ambient CO₂ (p < 0.001; SI Table 1), and the decrease in K_{MH} at 6 °C in high CO₂ was significantly different from that at 6 °C in ambient CO₂ (p < 0.01; SI Table 1).

Overall, fish in the 6 °C ambient CO₂ treatment started and remained in high condition throughout the entire experiment. The random effect of tank had minimal effect on K_{MH} (SI Table 1).

K_{DW} was significantly affected by interactions between temperature and DPH (X85 = 18.697, p < 0.001; Table 5), and CO₂ and DPH (X85 = 4.282, p < 0.05; Table 5), which reflected differing dynamics between temperature and CO₂ as the Pacific cod larvae aged (Fig. 5). At 0 DPH, fish incubated at 3 °C had significantly lower K_{DW} than at 6 (p < 0.05; SI Table 1) but not 10 °C (p > 0.05; SI Table 1). At 3 °C, as fish aged, K_{DW} remained constant and generally low (i.e. < 0) at ambient CO₂, but there was an increase in K_{DW} under high CO₂ conditions (slope = ambient: -0.0009 vs high: 0.0034; SI Table 1). At 6 °C, K_{DW} slightly increased as fish aged, and this improvement in condition was more prominent in the ambient CO₂ condition than in the high CO₂ condition (slope = ambient: 0.0017 vs high: 0.0006; SI Table 1). At 10 °C, both CO₂ treatments showed a significant increase in K_{DW} as fish aged (slope = ambient: 0.0173 vs high: 0.0182; SI Table 1). Among these relationships, the increase in K_{DW} with age at 3 °C high CO₂ and, separately, at 10 °C (both CO₂ treatments) was significantly different than the change in K_{DW} with age at 3 °C and ambient CO₂ (p < 0.05; SI Table 1). In addition, the less pronounced increase in K_{DW} at 6 °C in high CO₂ compared to at 6 °C in ambient CO₂ was also significantly different (p < 0.05; SI Table 1). Similar to K_{MH}, fish held at 6 °C in ambient CO₂ remained in higher condition throughout the experiment, and fish reared at 3 °C generally remained in lower condition.

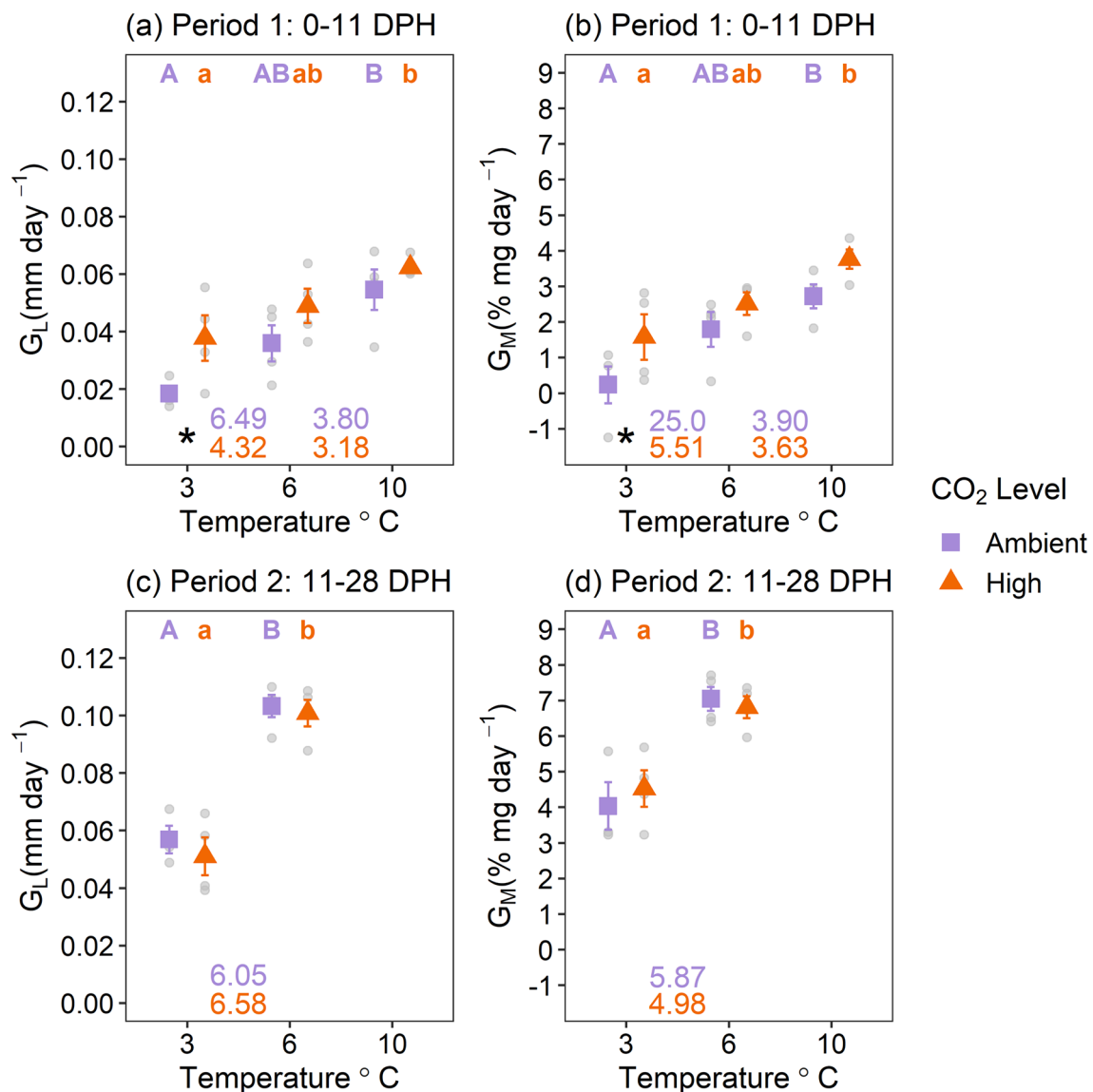


Fig. 3 Length-based (a, c) and mass-based growth rates (b, d) of Pacific cod across temperature with CO₂ level designated as ambient (purple square) and high (orange triangle). The age range of time period 1 (0–11 days post hatch [DPH]; a, b) includes all three temperature treatments while period 2 (11–28 DPH; c, d) does not include the 10 °C temperature treatment because this treatment ended at 11 DPH. Values are presented as mean ± S.E. and overlain atop the tank-level data (grey points; n=4 per treatment). Significant difference

between each temperature treatment within a specific CO₂ treatment are shown by different letters on the top. The * indicates a significant difference between CO₂ level within a temperature treatment. The numbers in between temperature treatments indicate the Q₁₀ value between each temperature interval (3–6; 6–10) shown for ambient and high CO₂ as the top value (purple) and bottom value (orange), respectively

Discussion

The eastern North Pacific Ocean ecosystem will continue to experience combined OW and OA, exposing resident fish species to these environmental stressors. Early life stages, which tend to be more sensitive, show varied responses to the combination of OW and OA. In our study, OW had a greater impact on embryonic and larval Pacific cod than OA, leading to very high mortality rates in fish incubated

at 10 °C. Across the measured responses, 6 °C appeared to be an optimal temperature for this species. Pacific cod responded to OA in some, but not all, measured traits and developmental stages. When there was a significant effect of OA, the response was generally positive (e.g., reduced embryonic daily mortality rates), but both condition indices were negatively impacted by OA at 6 °C. Under OW conditions (10 °C), there was no synergistic or additive effect when combined with OA. These results highlight the trait-,

Table 5 Analysis of variance (Wald-Chi square tests) of larval condition indices

	Df	χ^2	K_{MH}		K_{DW}	
				P-value	χ^2	P-value
Temperature	2	11.958		0.003	4.478	0.107
CO ₂	1	0.842		0.359	1.145	0.285
DPH	1	15.135		< 0.001	0.323	0.570
Temperature $\times$ CO ₂	2	0.747		0.688	1.048	0.592
Temperature $\times$ DPH	2	15.620		< 0.001	18.697	< 0.001
CO ₂ $\times$ DPH	1	1.772		0.183	4.220	0.040
Temperature $\times$ CO ₂ $\times$ DPH	2	9.050		0.011	4.282	0.118

Values in **bold** indicate a significant difference (p-value < 0.05). Full GLMM (K_{MH}) and GLM (K_{DW}) coefficients are in SI Table 1. K_{MH} = condition based on deviance in myotome height and length relationship. K_{DW} = condition based on deviance in dry weight and length

Fig. 4 The trend in morpho-metric-based condition (K_{MH}) as fish age (DPH) differed between temperature and CO₂ treatments. Predicted K_{MH} (line) is from the generalized linear-mixed effects model with shading representing the 95% prediction intervals. Trends are separated by temperature treatment (3, 6, 10 °C top to bottom) and CO₂ level (left column: ambient [purple]; right column: high [orange]). Grey points are values for individual fish data (n = 52–67 per treatment and DPH). The dashed line at 0 indicates the predicted myotome height based on the length of the fish. There was a significant effect of temperature, DPH, temperature $\times$ DPH, and temperature $\times$ DPH $\times$ CO₂

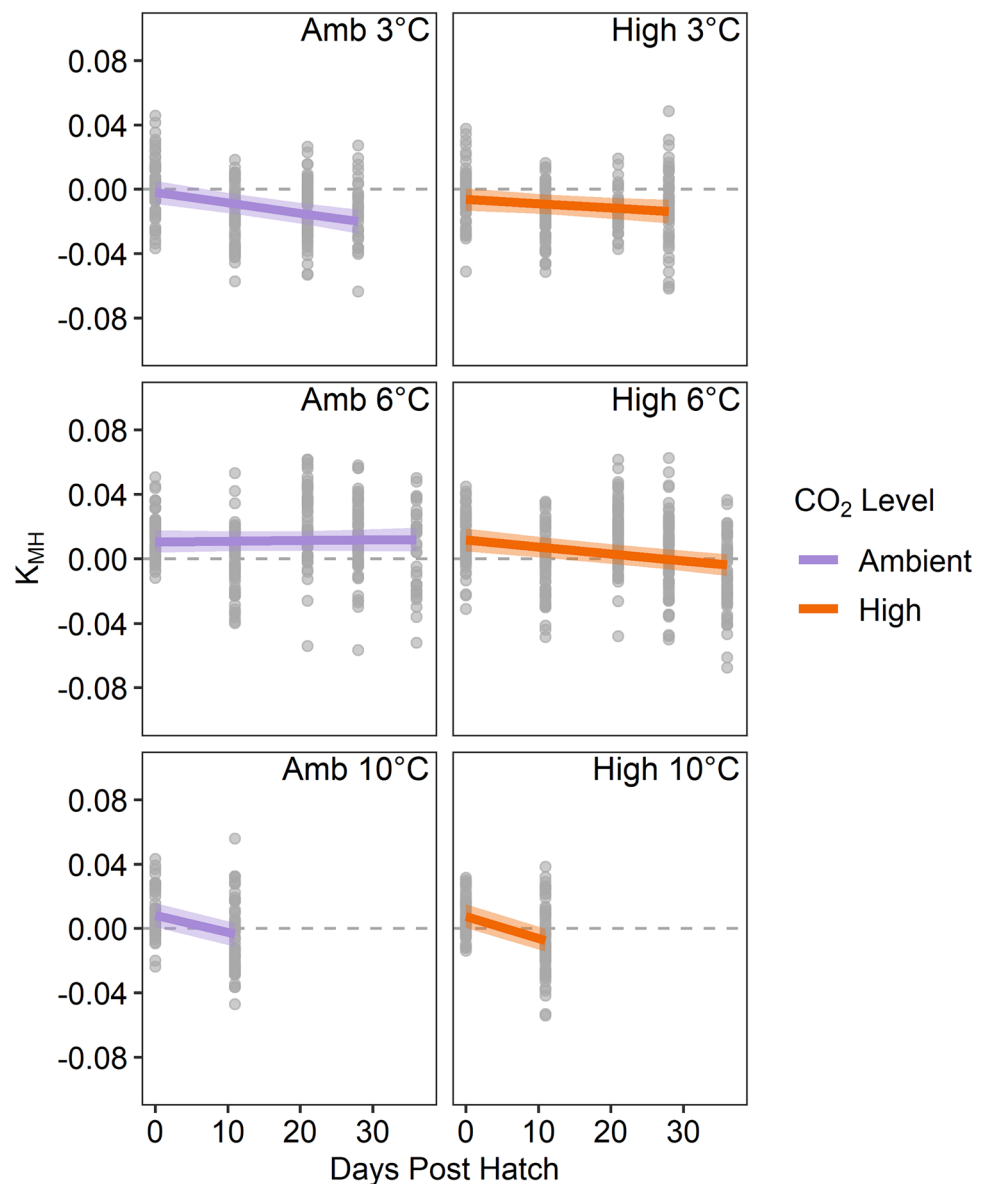
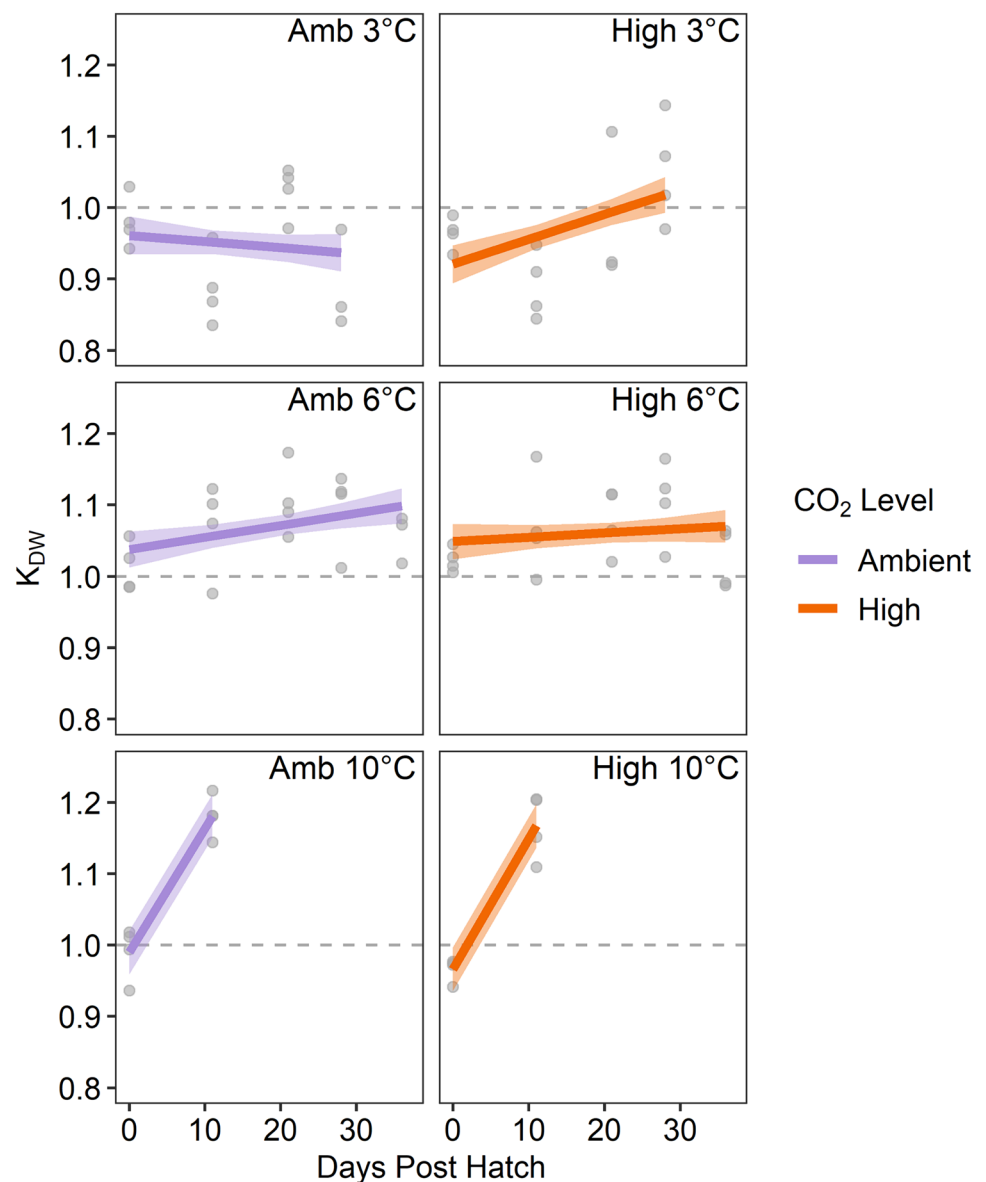


Fig. 5 The trend in weight-based condition (K_{DW}) as fish age (DPH) differed between temperature and CO_2 treatments. Predicted K_{DW} (line) is from the generalized linear model with shading representing the 95% confidence intervals. Trends are separated by temperature treatment (3, 6, 10 °C top to bottom) and CO_2 level (left column: ambient [purple]; right column: high [orange]). Grey points are values for tank-level data ($n=4$ per treatment and DPH). The dashed line at 1 indicates the predicted dry weight based on the length of the fish. There was a significant effect of temperature $\times$ DPH and $CO_2 \times$ DPH



stage- and species-specific responses of marine fish to multiple stressors.

Temperature affects all measured early life history traits

Across all measured traits, temperature was consistently a significant predictor for responses measured in both Pacific cod embryos and larvae. Within the temperature treatments explored here, 6 °C appeared to be most optimal for embryonic and larval stages. At 6 °C, Pacific cod had low mortality rates, larger yolk area at-hatch, and were consistently in the highest condition; at 10 °C, Pacific cod grew the fastest but were in poor condition with high daily mortality rates; and at 3 °C, Pacific cod daily mortality rates were lower but both growth and condition were also low.

The effects of temperature on Pacific cod in this study agree with previous studies on this species at the embryonic and larval life stages. Embryonic and larval daily mortality rates were lowest at 3 and 6 °C, which is similar to the optimal temperatures for hatch success reported in Bian et al. (2016) and Laurel and Rogers (2020). A narrow thermal window (~ 5 °C range) for successful hatch may signify that Pacific cod have evolved to develop in cold temperatures (Laurel and Rogers 2020). Pacific cod embryos reared at 10 °C continued to experience high mortality into the larval stage, which led to an early termination of the tanks maintained at 10 °C. Although in the wild Pacific cod embryos and larvae are separated vertically in the water column, recent warming events generated anomalously warm temperatures in both bottom and surface waters (Barbeaux et al. 2020),

potentially exposing Pacific cod to warmer than average temperatures throughout early life.

Pacific cod size-based traits were also temperature dependent, and followed well-established relationships between temperature and development. In this study, hatch size was largest at the lowest temperature and growth was fastest at the highest temperature, which support the findings in previous studies on Pacific cod size-at-hatch (Laurel et al. 2008) and growth (Laurel et al. 2011, 2016). Other gadid species have also shown similar relationships to size at hatch and growth with temperature, but the strength of the relationship is species-specific (Koenker et al. 2018; Laurel et al. 2018). Larger larval fish size is important as it can reduce size-dependent mortality (Anderson 1988), which would suggest lower temperatures would benefit larvae at hatch and higher temperatures would benefit larvae post-hatch. Condition indices can provide additional context to size data and illuminate other relationships to temperature. In both K_{MH} and K_{DW} , Pacific cod incubated at 6 °C hatched and remained in higher condition than those at 3 and 10 °C, which suggests that while the mid-temperature did not elicit maximum body sizes, it may have been more optimal for development.

In a warmer ocean, larval fish could benefit from faster growth (if food supply is adequate), but there may be additional physiological tradeoffs limiting fish at their upper thermal limits. Pacific cod in this study exhibited the fastest growth at 10 °C, but mortality rates were substantially higher in this temperature treatment. This result could have been driven by selective mortality (e.g. “characteristics of survivors”; Sogard 1997), and/or suggests impairment in additional physiological processes (i.e. metabolism; Gobler et al. 2018), which could increase the inefficiency in energy usage with growth (Murray & Klinger 2022). Altogether, the results in this study suggest prolonged exposure to warmer ocean conditions could increase mortality rates, and may lead to reduced recruitment to the adult population.

Responses to elevated CO₂ are trait, stage, and temperature dependent

In contrast to the pervasive influence of temperature, the responses to elevated CO₂ were not consistent across measured traits, life stages or temperature treatments. Most commonly, when there was a CO₂ effect, the response was positive. For example, elevated CO₂ led to lower embryonic daily mortality rates and faster growth from 0–11 DPH. At 6 and 10 °C, although not significant, yolk area was larger in elevated versus ambient CO₂ levels. The only negative effect of elevated CO₂ was elicited in both condition indices at 6 °C. Overall, these results highlight that the effects of OA are nuanced, and both the presence and directionality of an effect can be trait-, stage- and temperature-dependent.

The positive effects of OA included reduced embryonic mortality rates and faster growth rates of Pacific cod during an early stage of development (0–11 DPH). Other studies have also documented positive effects of OA on fish survival (Flynn et al. 2015) and growth rates (Munday et al. 2009; Hurst et al. 2013; Chambers et al. 2014; Schade et al. 2014; McCormick & Regish 2017). Potential causes of faster growth under elevated CO₂ could be attributed to increased metabolic rates and subsequent higher feeding rates (Ishimatsu et al. 2008; McCormick & Regish 2017) or driven by sensitivity of growth hormones to environmental stressors (Deane & Woo 2009; Bruzzio 2022). However, these positive effects of OA on Pacific cod at their earliest life stages (embryos) may reflect that their natal habitat is naturally higher in CO₂ than the ambient levels reported in our study. Pacific cod adults are demersal and spawn at a range of depths, where the embryos develop at ~50–250 m (Alderdice & Forrester 1971; Hirschberger & Smith 1983; Neidetcher et al. 2014). As such, both the adults and embryos are naturally exposed to a range of CO₂ levels that may be higher than ambient levels measured at the ocean surface (Pilcher et al. 2019; Hauri et al. 2023). Adult exposure can provide offspring with resiliency to OA as seen in larval white sea bass (*Atractoscion nobilis*; Kwan et al. 2021) and Atlantic cod (Stiasny et al. 2018). Benthic early life stages of fish have also been suggested to be resilient to OA (Cattano et al. 2018), reflecting the naturally higher CO₂ environment the fish develop in. As such, the better survival and growth under elevated CO₂ conditions may reflect that Pacific cod early life stages are adapted to these conditions.

After hatch, larval Pacific cod rise to the surface, and continue to develop in a pelagic surface environment (Hurst et al. 2009), where CO₂ levels are likely closer to the ambient levels we used in this study. Interestingly, depending on the measured trait, Pacific cod began to respond neutrally or negatively to OA at a later stage of development, which may coincide ontogenetically with this transition to developing in surface waters. Future work may benefit from designing experiments that set control conditions to mimic natural biogeochemical changes that occur throughout ontogeny to develop a more representative ambient treatment and assess the potential influence of static treatments used in many studies.

While there is evidence for positive OA effects, some fish species have been negatively affected by (Chambers et al. 2014; Pimentel et al. 2016; Baumann et al. 2022) or show no response to OA (Hurst et al. 2012, 2013, 2017; Perry et al. 2015). For larval Pacific cod in our study, the only negative effect of OA was a decrease in condition at the optimal temperature. Even though the type of response to OA may differ, many of these studies are similar such that when there is a significant response to OA it manifests in some but not all measured life stages and/or traits. For example,

sensitivity to OA was stage-specific, being present in early life stages for walleye pollock (*Gadus chalcogrammus*; Hurst et al. 2021) and in later stages for seabream (*Chrysoblephus laticeps*; Muller et al. 2020). In sand lance (*Ammodytes dubius*), OA sensitivity was exhibited in some but not all of the several measured traits (Baumann et al. 2022). Perhaps stage-specific responses are influenced by changes in the natural environment experienced throughout development, as it may for Pacific cod, and/or a reflection of a complicated interaction between development and changing physiological capacities.

Measurement endpoints in our and many other OA studies have focused on whole-animal responses including growth and mortality. These metrics are valuable for their ability to be more succinctly applied in stock assessments and used in broad understanding of global change biology. However, finer-scale physiological measurements can sometimes provide different insight into the consequence of high CO₂ exposure. For example, some studies have shown that OA increased growth rates, but at the cost of impaired organ, tissue, or skeletal development (Frommel et al. 2012; Chambers et al. 2014). In walleye pollock, OA did not affect larval growth rates, but swim-bladder inflation was negatively affected (Hurst et al. 2021). Hurst et al. (2019) also showed disrupted lipid metabolism in larval Pacific cod under OA conditions. Furthermore, from a subset of larval Pacific cod sampled in this study, fish incubated at 6 °C and elevated CO₂ downregulated β -hydroxyacyl CoA dehydrogenase activity, indicating a potential disruption in the fatty acid metabolism under OA conditions (E. Slesinger, unpub. data). This demonstrates that sensitivity to OA may manifest at finer physiological scales than reflected in the whole-animal responses examined here. Future studies may benefit from investigating the effects of OA on more sensitive measurement endpoints in an effort to establish a comprehensive assessment of sensitivity to elevated CO₂.

Many of the responses of Pacific cod to OA were temperature-dependent and not consistent between temperature treatments. Where there was an effect of elevated CO₂, the response tended to be positive at 3 °C, negative at 6 °C, and neutral at 10 °C. Temperature-dependent responses to OA have been found in multiple studies on early life stages of fish. Some multistressor studies demonstrated synergistic effects of OW and OA on larval fish where the combined effect of the two stressors led to a stronger response than at the level of the single stressor (Dahlke et al. 2017; Gobler et al. 2018; Villalobos et al. 2020). Other fish, such as larval Atlantic herring (*Clupea harengus*), were negatively impacted by high CO₂ at the colder temperatures but not affected at warmer temperatures (Sswat et al. 2018), whereas other species exhibited a consistent response to OA across multiple temperature treatments (Pimentel et al. 2016). Altogether,

this temperature-dependent response is notable because it suggests Pacific cod may be impaired by OA when held at optimal temperatures, but that there is no additional negative effect from OA when Pacific cod are exposed to OW. Therefore, some species may exhibit a matrix of responses across a temperature x OA landscape, which is important to consider when projecting potential population impacts under varying climate change scenarios.

Influences on species sensitivities to OA

Results in this study differed from the previous and only other study of OA effects on Pacific cod larvae (Hurst et al. 2019). Hurst et al. (2019) found a negative effect of OA on growth at 2 weeks post-hatch and a positive effect at 5 weeks post-hatch. In contrast, Pacific cod in this study were positively affected by OA at ~2 weeks post-hatch, and there was no effect of OA at 5 weeks post-hatch. In addition to differences in the direction of the response to OA, the two studies also differed in general trends in size and growth. Overall, there was slower growth in this study within both CO₂ treatments and life stages when compared to growth in Hurst et al. (2019), which may have been driven by slightly lower incubation temperatures (6 °C this study; 7.4 °C Hurst et al. (2019)). In this study, CO₂ exposure was initiated 72 h after fertilization while in Hurst et al. (2019), embryos were incubated at ambient CO₂ and exposed to elevated CO₂ levels starting at 3 DPH. For Pacific cod in this study, slower growth in the elevated CO₂ treatment compared to fish in Hurst et al. (2019) may have been a consequence of prolonged CO₂ exposure, but does not explain slower growth between the ambient CO₂ treatments across the two studies. Broodstock origin may have also contributed to the observed differences. In this study, wild fish were used to produce gametes while Hurst et al. (2019) sourced their larvae from embryos produced by laboratory broodstock that had been maintained at Alaska Fisheries Science Center's laboratory in Newport, OR, USA. While both parental groups were originally sourced from coastal waters off Kodiak Island, AK, USA, the adults in Hurst et al. (2019) had been held in the laboratory facilities for several years prior to their use in the experiment. Parental quality and/or environmental history immediately prior to the spawning season could have contributed to the differing larval sensitivities seen between the two studies. Despite differences in methods and specific responses, the results are consistent in demonstrating stage-specific responses to elevated CO₂ and that the growth effects of OA are modest – in comparison to thermal effects – and most apparent during the earliest stages of development.

The limitations of collecting and shipping wild adult Pacific cod restricted us to use the eggs from one female for this study. As such, the results here may not represent the entire population-wide responses of Pacific cod sourced from multiple parental

lines. Maternal affects can lead to changes in the response of offspring to environmental stressors (Stiasny et al. 2018), and as such, these results may not represent the entire population. The use of wild broodstock-sourced embryos can mitigate potential confounding factors associated with fish in routine laboratory husbandry. For example, laboratory vs wild-caught fish can differ in lipid storage (Copeman et al. 2020). Future studies would benefit by testing more wild-caught samples to confirm the generality of the response observed here.

Pacific cod embryos and larvae were not under food limitation throughout the duration of the experiment. Other studies have shown that OA resiliency wanes under low food quantity or quality (Gobler et al. 2018; Stiasny et al. 2018). Food availability is also important to consider in the context of ocean warming, where the mortality rates at 10 °C could be even higher under food limitation (Koenker et al. 2018). In some cases, feeding treatment can be a dominant driver in multistressor experiments (Hurst et al. 2017, 2019). Interactions between OW, OA, and food are nontrivial as food can influence Pacific cod survival under different warming scenarios (Copeman and Laurel 2010; Laurel et al. 2011, 2021), and prey items in the environment may themselves be more sensitive to OA (Fabry et al. 2009; Melzner et al. 2009).

Conclusions

Results presented here suggest that Pacific cod embryos and larvae were more sensitive to OW than OA. These results are consistent with the findings from other OW-OA multistressor studies on marine fish early life stages (Leo et al. 2018; Cominassi et al. 2019). Of the few instances where there was an effect of OA at a specific temperature, the duration of the response was stage-specific and ephemeral, rarely lasting the full experimental timeframe. Positive effects of OA at the earliest life stages (embryos) may reflect the naturally higher CO₂ levels of benthic environments of the Gulf of Alaska and Bering Sea. These positive effects waned as fish aged, which also coincides with the transition of larvae inhabiting surface waters post-hatch. The single instance of a negative effect of OA reported here occurred at 6 °C, which is near optimal temperatures for this species. Importantly, Pacific cod did not exhibit synergistic or additive responses to combined OW and OA, suggesting OW is likely a more prominent threat in future ocean conditions. More sensitive metrics that focus on metabolism, development, or energetics, may illuminate additional responses to OA that are not evident in whole-animal metrics measured here. Continued research on the combined effects of OW and OA on Pacific cod will be important for the improvement of predictions under climate change scenarios necessary for future management of this commercially important species.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00227-024-04439-w>.

Acknowledgements We thank Paul Iseri and Chris Magel for maintenance of the pH and temperature systems used in this experiment. We acknowledge Jessica Andrade, Carly Salant, and Mara Spencer for their assistance in fish culture. Michele Ottmar and Mary Beth Rew Hicks helped with general laboratory maintenance. We are also grateful to Alissa Abookire and Captain Nate Rose for providing Pacific cod embryos from Kodiak Island, AK. Joe Langan, Jessica Miller and Zoe Almeida provided valuable discussion on statistical analyses, and Zoe Almeida and Chris Chambers provided constructive feedback on an earlier draft of this manuscript. Reference to trade names does not imply endorsement by the National Marine Fisheries Service. The findings and conclusions in this paper are those of the authors and do not necessarily represent the views of the National Marine Fisheries Service.

Author contributions ES, TPH, and BJL conceptualized and designed the experiment. Data collection was performed by ES, SM and TPH, and data analysis was performed by ES. ES led the writing of the manuscript with contributions by SM, TPH and BL. All authors reviewed and contributed to the writing of the final manuscript.

Funding This work was supported by grant #20903 to TPH from NOAA's Ocean Acidification Program (Research Organization Registry 02bf4816).

Ocean Acidification Program, 20903, Thomas P Hurst

Data availability All data from this project will be made publically available through NOAA's National Centers for Environmental Information at https://www.nodc.noaa.gov/oads/stewardship/data_assets.html.

Declarations

Conflicts of interest The authors have no conflicts of interest.

Ethical approval This research was carried out in accordance with all applicable institutional and national guidelines. Pacific cod embryos were collected under Alaska Department of Fish and Game Permit #P-22-005. This research was conducted at NOAA's Alaska Fisheries Science Center Laboratory in Newport, Oregon. NOAA National Marine Fisheries Service does not have an Institutional Animal Care and Use Committee (IACUC) approval process for research on fishes. All work followed American Fisheries Society policies on the Guidelines for Use of Fishes in Research (American Fisheries Society: https://fisheries.org/docs/policy_useoffishes.pdf) and the AVMA Guidelines on Euthanasia (American Veterinary Medical Association: <https://olaw.nih.gov/sites/default/files/Euthanasia2007.pdf>).

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